

The Impact of Public Assistance Spending, Education Rates, and Cost of Living on State's Poverty Rates

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This analysis examines the relationship between public assistance spending per capita, education rate, cost of living index, and poverty rates in the United States at the state level. The goal is to determine whether states that spend more on public assistance programs, have higher education rates, or face higher cost of living tend to have lower poverty rates.

Data Definitions

Poverty: The US Census compares the income sources of a household (including, earnings, unemployment compensation, workers' compensation, Social Security, Supplemental Security Income, Public assistance, Veterans' payments, Survivor benefits, Pension or retirement income, etc), to create poverty thresholds, which varies based on the size and ages of the household. <https://www.census.gov/topics/income-poverty/poverty/guidance/poverty-measures.html>

Spending: Public welfare spending in particular, by state. Public welfare expenditures include cash assistance through Temporary Assistance for Needy Families (TANF), Supplemental Security Income, and other payments made directly to individuals, as well as payments to physicians and other service providers under programs like Medicaid, which can be listed under 'hospitals' in budgets. <https://www.census.gov/data/datasets/2022/econ/local/public-use-datasets.html>

Education Rates: Educational attainment, as defined by the US Census, is the highest degree or level of school a person has completed. <https://www.census.gov/topics/education/educational-attainment/about.html>

Cost of Living (coli): Cost of living is the amount of money needed to sustain a certain standard of living by affording expenses such as housing, food, taxes, and healthcare, with the Index developed by the Council for Community and Economic Research (C2ER). <https://www.coli.org/about/what-is-coli/>

Uploading and Cleaning Data

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidycensus)
library(readr)
library(knitr)
library(dplyr)
library(stringr)

#Uploading poverty data from Census Bureau, through the American Community
#Survey, through tidycensus API
poverty_data <- get_acs(
  geography = "state",
  variables = "B17001_002",
  year = 2022,
  survey = "acs5"
)
```

```
## Getting data from the 2018-2022 5-year ACS
```

```
poverty_data_clean <- poverty_data %>%
  select(state = NAME, poverty_rate = estimate)
acs_vars <- load_variables(2022, "acs5", cache = TRUE)

poverty_data_clean <- poverty_data_clean %>%
  mutate(state = tolower(state))
poverty_data_clean <- poverty_data_clean %>%
  filter(state != "puerto rico")

#Uploading spending data, from US Summary and State Estimate Tables, from 2022
#State & Local Government Finance Historical Datasets and Tables
spending_data <- read.csv("/Users/k.churchill22/Library/CloudStorage/OneDrive-SaintLouisUniversity/socialsci/2022/2022_Summary_and_State_Estimate_Tables/2022_Summary_and_State_Estimate_Tables.csv")
spending_data_clean <- spending_data %>%
  pivot_longer(cols = -Description, names_to="State", values_to="Spending") %>%
  filter(Description == "Public Welfare") %>%
  select(State, Spending)
spending_data_clean$State <- gsub(" ", ".", spending_data_clean$State)

spending_data_clean <- spending_data_clean %>%
  mutate(
    State = str_replace_all(State, "\\.", " "),
    Spending = as.numeric(str_replace_all(Spending, ",", ""))
  ) %>%
  filter(State != "United States Total")

spending_data_clean <- spending_data_clean %>%
  rename(state = State) %>%
  mutate(state = tolower(state))

spending_data_clean <- spending_data_clean %>%
  mutate(state = tolower(state))
```

Additional Data Considerations

```
#Uploading education data, through tidycensus API
education_data <- get_acs(
  geography = "state",
  variables = c(bachelors = "B15003_022",
                masters = "B15003_023",
                professional = "B15003_024",
                doctorate = "B15003_025",
                total_population = "B15003_001"),
  survey = "acs5",
  year = 2022
)
```

Getting data from the 2018-2022 5-year ACS

```
education_clean <- education_data %>%
  group_by(NAME) %>%
  summarise(
    educated_population = sum(estimate[variable %in% c("bachelors", "masters", "professional", "doctorate")],
    total_population = sum(estimate[variable == "total_population"]),
    education_rate = (educated_population / total_population) * 100,
    .groups = "drop"
  ) %>%
  select(state = NAME, education_rate)
education_clean$state <- tolower(education_clean$state)

#Uploading cost of living data, from C2ER data
coli_data <- read.csv("/Users/k.churchill22/Library/CloudStorage/OneDrive-SaintLouisUniversity/cosli_data.csv")
coli_data <- coli_data %>%
  select(state, Index)
coli_data <- coli_data %>%
  mutate(state = tolower(state))
coli_data <- coli_data %>%
  filter(!is.na(Index))
coli_data$state <- trimws(tolower(coli_data$state))
coli_data$state <- trimws(coli_data$state)
coli_data$state <- gsub("\\s+$", "", coli_data$state)

poverty_data_clean$state <- str_trim(poverty_data_clean$state)
coli_data$state <- str_trim(coli_data$state)

poverty_data_clean$state <- tolower(poverty_data_clean$state)
coli_data$state <- tolower(coli_data$state)
```

Completing Regression Analysis

```
# Merge the datasets on 'state'
final_data <- poverty_data_clean %>%
  left_join(spending_data_clean, by = "state") %>%
  left_join(education_clean, by = "state") %>%
```

```
left_join(coli_data, by = "state")

model <- lm(poverty_rate ~ Spending + education_rate + Index, data = final_data)
summary(model)
```

```
##
## Call:
## lm(formula = poverty_rate ~ Spending + education_rate + Index,
##     data = final_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -599113 -182178  -91755   85453 1715323
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.113e+06  3.314e+05   3.359  0.00156 **
## Spending       3.297e-02  2.114e-03  15.598 < 2e-16 ***
## education_rate -4.243e+03  9.807e+03  -0.433  0.66727
## Index         -7.749e+03  3.600e+03  -2.152  0.03653 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 384800 on 47 degrees of freedom
## Multiple R-squared:  0.8401, Adjusted R-squared:  0.8299
## F-statistic: 82.31 on 3 and 47 DF,  p-value: < 2.2e-16
```

Interpretations

This analysis provides insights into the relationship between state-level public assistance spending, education rates, cost of living, and poverty rates. Spending has a strong positive association with poverty rate, which suggests that states with higher public assistance spending tend to have higher poverty rates, although this relationship does not necessarily imply causality, as states with more poverty may simply spend more in response to need.

The cost of living index is negatively associated with poverty rates, suggesting that states with a higher cost of living tend to have lower reported poverty rates, with a statistically significant p-value of 0.0365. However, this counterintuitive result necessitates further clarification, particularly in relation to wages, social benefits, and the purchasing power between states.

The lack of statistical significance (p-value = 0.66727) for education rates raises questions about the timescale of education's impact on poverty reduction and whether additional factors, such as labor market conditions, mediate this relationship. Future research could explore the long-term effects of education on poverty using different models or longitudinal data, compared with earning potentials between different industries.

While this model captures broad trends, future refinements should address potential confounding variables, multicollinearity, and causal inference techniques to strengthen the analysis. Further investigation could also explore whether specific types of public assistance spending (e.g., direct cash transfers vs. food assistance) have different effects on poverty reduction.

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